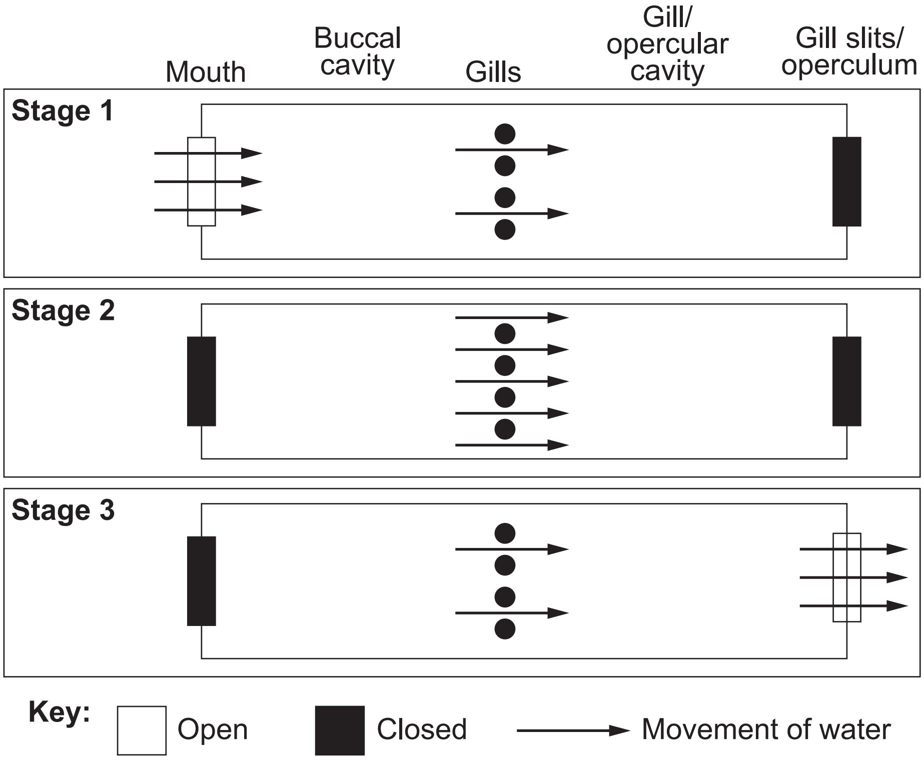
Examiner only

5. Bony fish use a ventilation mechanism to move water over their gills. **Image 5.1** shows three stages during ventilation.

Image 5.1



(a) Use **Image 5.1** and your own knowledge to explain how water is moved across the gills. [4]

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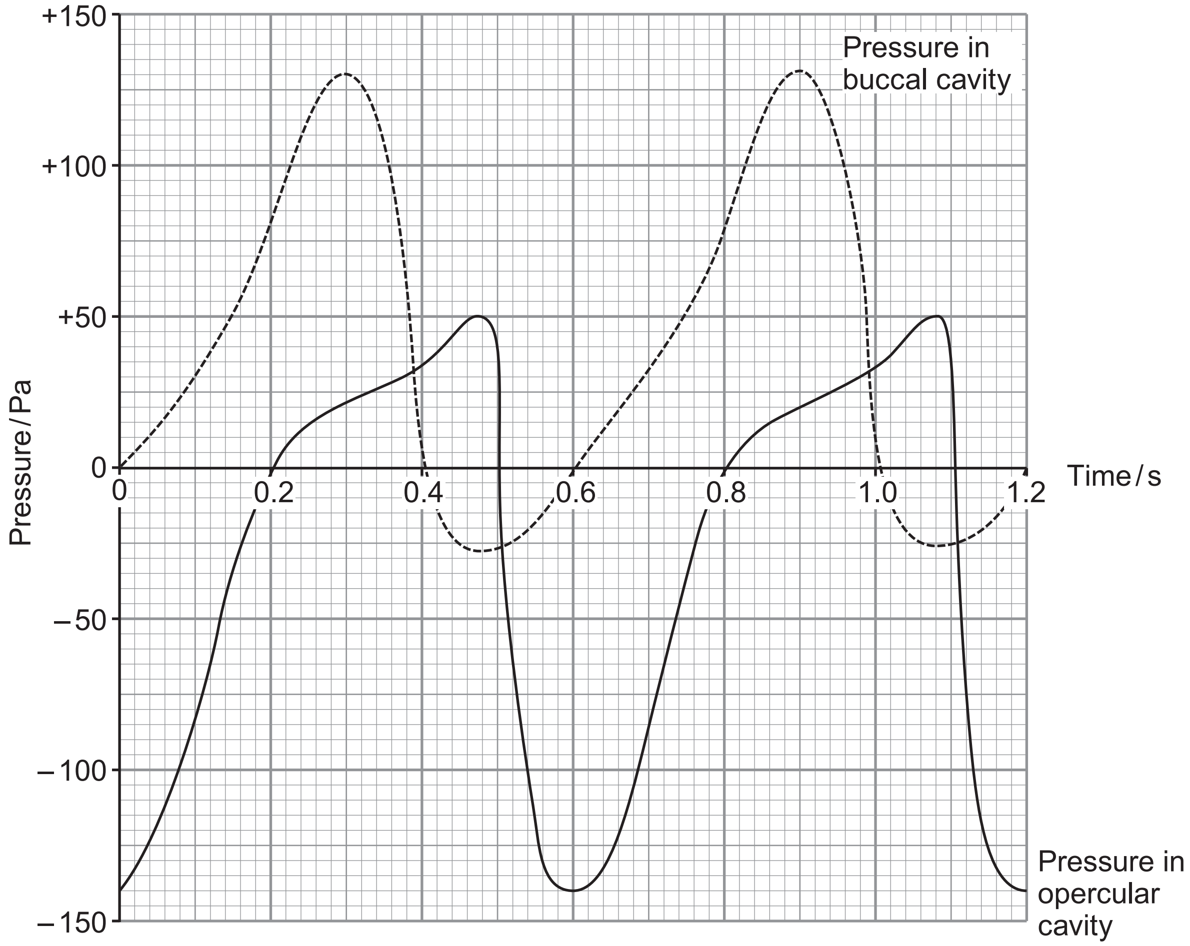
(b) **Graph 5.2** shows the pressure changes in the buccal and opercular cavities during two ventilation cycles.

Graph 5.2



(i) Use **Graph 5.2** to calculate the maximum pressure change within the **opercular cavity**. [1]

Maximum pressure change = ... Pa

(ii) I. Indicate **on Graph 5.2, with an arrow** labelled **B**, a point when water will enter the buccal cavity. [1]

II. Indicate **on Graph 5.2, with an arrow** labelled **G**, a point when water will be flowing over the gills. [1]

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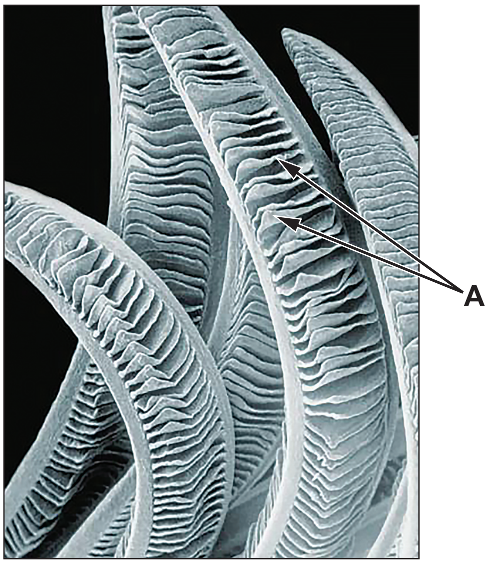
(c) **Image 5.3** is a magnified image of part of a fish's gill.

Image 5.3



(i) State the name of the structures labelled **A** in **Image 5.3**. [1]

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(ii) Use **Image 5.3** to describe and explain **one** adaptation of the gills for gas exchange. [2]

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(iii) The blood flow through the structures labelled **A** is in the opposite direction from the water flowing over the gills. Explain the advantage of this to the fish. [3]

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(d) The Australian Lungfish *(Neoceratodus forsteri)* is one of several species of fish that have evolved simple lungs. The lungs are subdivided into numerous smaller air sacs surrounded by many capillaries. Lungfish are capable of carrying out gas exchange via their gills and lungs.

During the dry season, water levels fall, and the water temperature rises. The concentration of oxygen in the water decreases. During these months the lungfish gulp air from the surface.

Suggest how these adaptations have enabled the lungfish to survive in its environment. [2]

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(e) The structure of the lungs of lungfish are homologous to the lungs of mammals. State what conclusion can be made about the evolutionary relationships between lungfish and mammals. [1]

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